

Effects of a Topically Applied Bioadhesive Berry Gel on Loss of Heterozygosity Indices in Premalignant Oral Lesions

Purpose The aim of this study was to assess the effects of topical application of a 10% (w/w) freeze-dried black raspberry (FBR) gel on oral intraepithelial neoplasia (IEN) variables that included histologic diagnoses and loss of heterozygosity (LOH) indices. Microsatellite instability and/or LOH at tumor suppressor gene – associated chromosomal loci have been associated with a higher risk for oral IEN progression to oral squamous cell carcinoma. Previously, our laboratories have shown that FBRs are well tolerated and possess potent antioxidant, apoptotic, and differentiation-inducing properties. **Experimental Design** Each participant with IEN served as their own internal control. Before treatment, all lesions were photographed, and lesional tissue was hemisected to obtain a pretreatment diagnosis and baseline biochemical and molecular variables. Gel dosing (0.5 g applied four times daily for 6 weeks) was initiated 1 week after the initial biopsy. Genomic DNA was isolated from laser-captured basilar and suprabasilar surface epithelial cells followed by PCR amplification using primer sets that targeted known and presumed tumor suppressor gene loci associated with INK4a/ARF, p53, and FHIT. Allelic imbalance was determined by sequence analysis using normal participant tissues to establish microsatellite marker peak patterns and allele sizes. **Results** Confirming earlier phase I data, none of the 27 participants developed FBR gel – associated toxicities. Furthermore, our results show histologic regression in a subset of patients as well as statistically significant reduction in LOH at tumor suppressor gene – associated loci. **Conclusions** These preliminary data suggest that further evaluation of berry gels for oral IEN chemoprevention is warranted.